PAX COUNTER

IoT-Based Footfall & Proximity Detector

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IoT Project (BCSE401L) Project by:

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OVERVIEW

PAX COUNTER is a **low-cost IoT** system that estimates **crowd presence** using an **ESP32** in Wi-Fi **promiscuous** mode. It passively detects Wi-Fi **probe requests** from nearby smartphones to count **unique devices** without requiring any **user interaction** or **network connection**.

Signal strength (**RSSI**) is used to approximate **proximity**, and all processing happens **on the device** itself. The results are formatted as **JSON** and **streamed** to a **dashboard** in **real time**, enabling **privacy-conscious occupancy** and **crowd analytics**.

OBJECTIVE

To design a **passive**, **low-cost** crowd monitoring system using an **ESP32** operating in Wi-Fi **promiscuous** mode, capable of detecting **probe request** packets from **nearby** smartphones without requiring active connections or user participation.

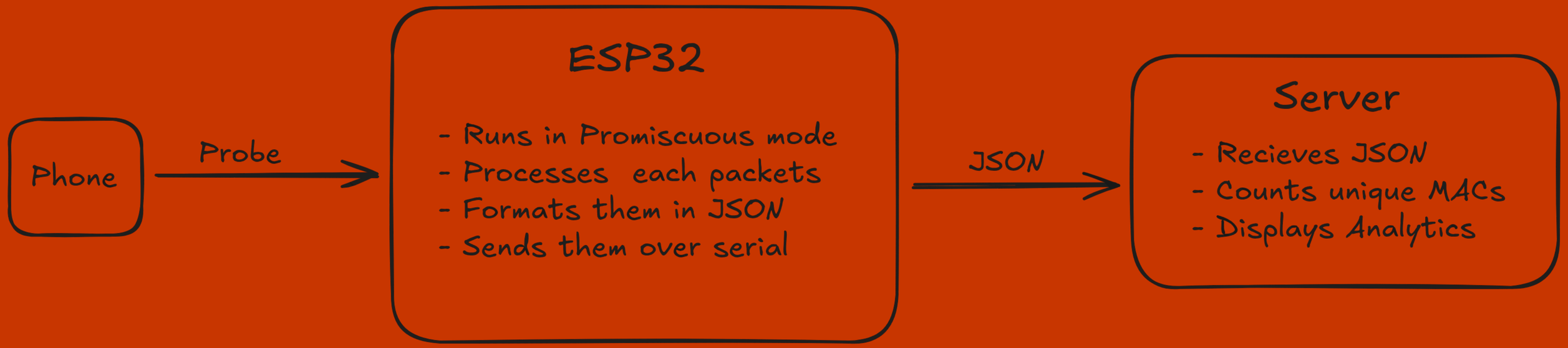
To process captured packets on-device by filtering duplicates, counting unique devices, estimating proximity using RSSI, and transmitting real-time metrics as structured JSON data to a dashboard for live visualization.

INTRODUCTION

With more connected devices around us, it’s possible to understand how crowded a place is using wireless signals instead of cameras. This project focuses on counting how many people are present in an area in a simple and low-cost way.

The system uses an ESP32 to listen for Wi-Fi signals sent out by smartphones and uses their strength to guess how close they are. This allows basic crowd and proximity tracking while avoiding privacy issues caused by cameras or user accounts.

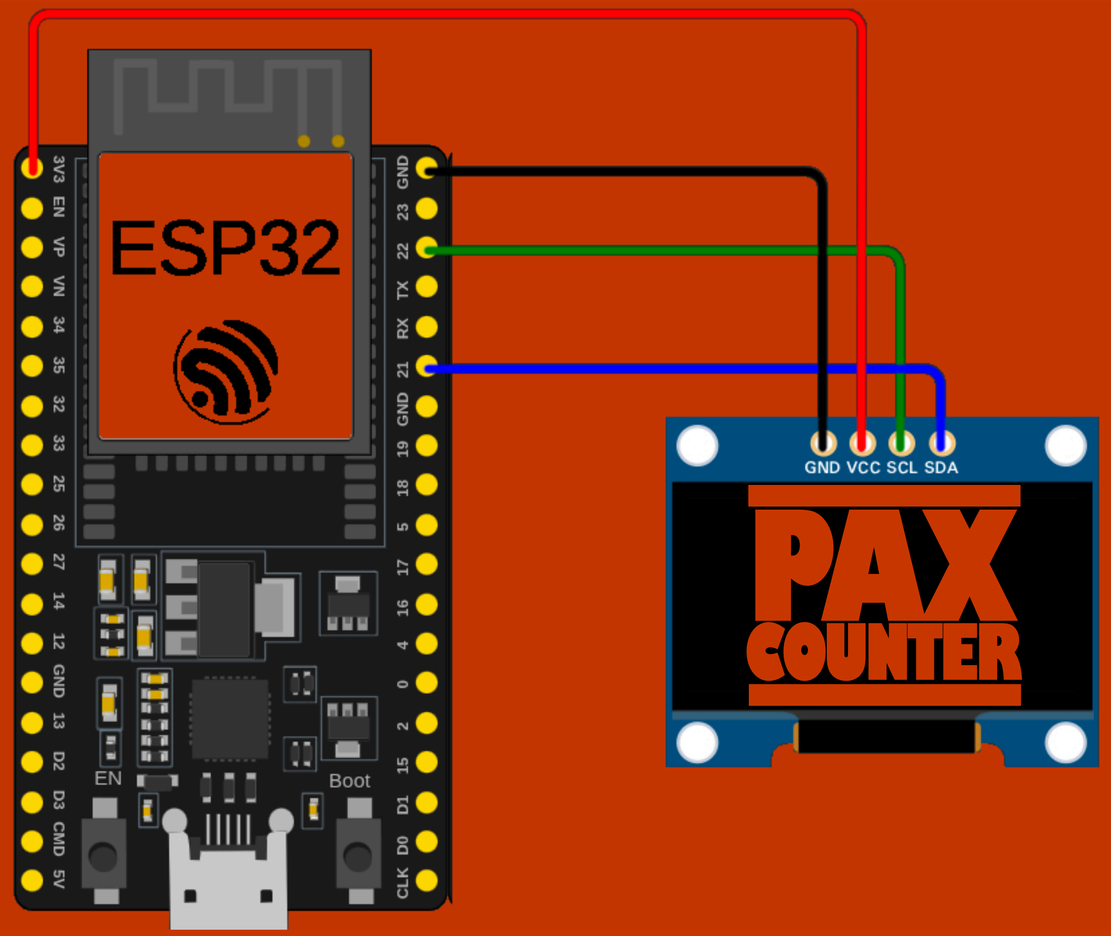
ARCHITECTURE



COMPONENTS

ESP32 Dev Board
OLED Display
USB Type-C Cable
Laptop / PC

CONNECTIVITY DIAGRAM



NOVELTY

The system estimates crowd presence without using cameras or asking users to install or connect to anything.

Along with counting devices, it also gives a rough idea of how close they are, using only signal strength and on-device processing.

RELATED WORK

1. Line-of-sight people counters (2021):
These systems count people entering or exiting an area but fail in open spaces where movement is unstructured.

2. Wi-Fi-based occupancy using existing infrastructure (2022):
Uses campus or building Wi-Fi networks and requires active device connections, increasing cost and setup complexity.

3. Passive Wi-Fi crowd analysis with backend processing (2020):
Detects devices passively but relies on servers or cloud processing instead of lightweight edge computation.

4. UWB-based localization systems (2020):
Offers high accuracy but depends on expensive hardware, making it unsuitable for low-cost crowd monitoring.

REFERENCES

1. Adisorn Sirikham et al., "Development of Bi-Directional Portable People Counter System based IoT," 2021.

2. Iresha Pasquel Mohottige et al., "Modeling Classroom Occupancy Using Data of WiFi Infrastructure in a University Campus," 2022.

3. Yuren Zhou et al., "Understanding Crowd Behaviors in a Social Event by Passive WiFi Sensing and Data Mining," 2020.

4. Vitomir Djaja-Josko, "Novel method for the wireless synchronization of the anchors in the UWB localization system," 2020.